

Chapter 19: Cost Estimation

Learning Objectives

By the end of this chapter, you will:

- Understand the major cost drivers of enterprise AI solutions.
 - Learn how to estimate and monitor AI program budgets across the product lifecycle.
 - Develop practical frameworks for forecasting cloud, model, and operational expenses.
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Beginner Explanation

One of the fastest ways to lose executive support for an AI program is to underestimate costs.

Unlike traditional applications, AI systems have variable costs based on:

- Number of users
- Token consumption
- Storage requirements
- Search queries
- Infrastructure usage
- Monitoring and support

Think of AI costs like an electricity bill:

The more you use it, the more you pay.

Example

Suppose:

- 10,000 users
- 20 requests per day
- 2,000 tokens per request

Monthly usage could quickly become millions of tokens.

Typical AI Cost Components

Component	Example
Model Usage	Azure OpenAI
Search	Azure AI Search
Storage	Blob Storage
Compute	Azure Functions

Component	Example
Monitoring	Azure Monitor
Networking	Front Door and API Management

AI TPMs must understand these costs to avoid surprises.

Intermediate Explanation

Enterprise AI budgeting typically includes three categories.

Capital Expenses (CAPEX)

- Initial implementation
- Training
- Consulting
- Infrastructure setup

Operational Expenses (OPEX)

- Model consumption
- Search queries
- Storage
- Monitoring
- Support

Hidden Costs

- Governance
- Compliance reviews
- User training
- Change management
- Security audits

Cost Estimation Framework

Category	Questions
Users	How many users will access the system?
Requests	How many requests per day?
Tokens	Average tokens per request?
Availability	SLA requirements?
Regions	Single or multi-region deployment?
Growth	Expected user growth?

Common Cost Drivers

Service	Cost Driver
Azure OpenAI	Tokens
Azure AI Search	Queries and indexes
Azure Functions	Executions
Storage	Data volume
Monitoring	Log ingestion
Networking	Traffic volume

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures solutions remain financially sustainable.
Risk	Identifies potential cost overruns early.
Stakeholder Management	Provides accurate forecasts to executives and finance teams.
Budget	Tracks actual spend versus approved budgets.

Enterprise Example

Business Problem

A global enterprise planned to deploy an AI assistant to 100,000 employees.

Initial estimates assumed:

- Low usage
- Minimal infrastructure costs
- Limited support requirements

After pilot deployment, actual costs were significantly higher due to increased adoption.

Solution

Implement a formal AI Cost Management framework.

Architecture

- Azure OpenAI
- Azure AI Search
- Azure Functions

- Azure Monitor
- Power BI
- Azure Cost Management

Outcome

Metric	Result
Budget Accuracy	Improved from 65% to 94%
Cost Variance	Reduced by 70%
Executive Confidence	Increased significantly
ROI	Achieved within 12 months

Lessons Learned

- Adoption drives costs.
 - Token monitoring is essential.
 - Pilot programs improve forecasting accuracy.
 - Cost dashboards should be reviewed weekly.
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Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	Model consumption
Azure AI Search	Enterprise retrieval
Azure Functions	Workflow execution
Azure AI Foundry	Usage and evaluation tracking

Supporting Services

- Azure Cost Management
 - Power BI
 - Azure Monitor
 - Application Insights
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Implementation Workflow

Step 1: Estimate Usage

Document:

- Number of users

- Daily requests
 - Average tokens
 - Growth assumptions
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Step 2: Calculate Monthly Costs

Example:

Metric	Value
Users	10,000
Requests per Day	20
Tokens per Request	2,000
Monthly Tokens	1.2 Billion

Step 3: Estimate Supporting Costs

Include:

- Search
 - Storage
 - Monitoring
 - Networking
 - Support
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Step 4: Implement Monitoring

Track:

- Daily spend
 - Monthly forecasts
 - Budget variance
 - Cost trends
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Step 5: Optimize

Examples:

- Smaller models
 - Caching
 - Better prompts
 - Search optimization
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Mermaid Diagram

```
```mermaid id="yj5jvs" graph TD
```

```
A[User Requests]
```

```
A --> B[Azure OpenAI]
```

```
B --> C[Azure AI Search]
```

```
C --> D[Azure Functions]
```

```
D --> E[Azure Cost Management]
```

```
E --> F[Power BI Dashboard]
```

```
F --> G[AI TPM]
```

```

```

```
Cost Estimation Model
```

```
```mermaid id="kl9j9z"
graph LR
```

```
A[Users]
```

```
A --> B[Requests]
```

```
B --> C[Tokens]
```

```
C --> D[Azure Costs]
```

```
D --> E[Monthly Budget]
```

```
E --> F[Executive Review]
```

TPM Checklist

- ☐ Estimate user volume.
- ☐ Forecast token consumption.
- ☐ Include operational costs.
- ☐ Configure budget alerts.
- ☐ Create cost dashboards.
- ☐ Conduct monthly reviews.
- ☐ Validate ROI assumptions.
- ☐ Optimize infrastructure.

- [] Review vendor pricing.
- [] Present budget updates.

Templates

Cost Estimation Worksheet

Component	Monthly Cost
Azure OpenAI	
Azure AI Search	
Azure Functions	
Monitoring	
Storage	
Total	

Monthly Cost Dashboard

KPI	Target
Budget Variance	<5%
Cost per User	Within Target
ROI	Positive
Token Usage	Within Forecast
Availability	99.9%

Risk Register

Risk	Impact	Mitigation
Cost Overruns	High	Budget alerts
Unexpected Adoption	Medium	Capacity planning
Vendor Price Changes	Medium	Quarterly reviews
Inefficient Prompts	Medium	Prompt optimization

Common Mistakes

1. Ignoring token consumption.
 2. Excluding operational expenses.
 3. Underestimating user adoption.
 4. Neglecting monitoring costs.
 5. Assuming costs remain constant.
 6. Failing to establish budget alerts.
 7. Not validating assumptions through pilot programs.
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Key Takeaways

- AI costs are highly variable and must be actively managed.
 - Token consumption is often the largest cost driver for AI applications.
 - Pilot programs improve forecasting accuracy.
 - Azure Cost Management and Power BI provide visibility into AI spending.
 - AI TPMs should review costs regularly and communicate budget impacts proactively.
 - Financial discipline is a key differentiator between successful and unsuccessful AI programs.
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Footer
